

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. II Semester Examination, 2025

Regulation: R20 | University: JNTUK

Machine Learning (21CS61)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Regression]

- a) Explain gradient descent and its variants.
- b) Derive the normal equation for linear regression.

UNIT - 2 [Classification & Model Evaluation]

- a) Explain logistic regression with the sigmoid function.
- b) Explain the SVM algorithm with the kernel trick.

UNIT - 3 [Unsupervised Learning & Dimensionality Reduction]

- a) Explain k means clustering with an example.
- b) Explain PCA and its role in dimensionality reduction.

UNIT - 4 [Neural Networks & Deep Learning]

- a) Explain backpropagation with a neat diagram.
- b) Explain backpropagation with a neat diagram.

UNIT - 5 [Ensemble Methods & Model Selection]

- a) Compare bagging and boosting techniques.
- b) Compare bagging and boosting techniques.

Backlog Buddy - previous question paper library. This paper is reconstructed from previous examination papers available on the platform and is intended for practice only.